Architectural Blueprint: Strict Replication of GGR (2006)

Correcting Implementation Deviations for Borsa Istanbul (2010–2026)

Objective

The target Language Model must rewrite the existing pairs trading codebase to strictly align with the classic relative-value arbitrage rule defined by Gatev, Goetzmann, and Rouwenhorst (2006). The current implementation contains fundamental structural deviations from the paper that must be explicitly corrected.

Part 1: Critical Deviations to Correct

The provided source files (`notebook.ipynb`, `strategies.py`, `indicators.py`, etc.) diverge from the strict academic methodology of GGR (2006) in the following ways:

1. **Pre-selected vs. Exhaustive Search:** The current code relies on a predefined list of 48 intra-sector pairs (`CANDIDATE_PAIRS`). GGR (2006) is sector-agnostic. It exhaustively computes the Sum of Squared Deviations (SSD) across *all* available liquid equities in the market, selecting the Top 20 pairs blindly based on distance.

2. **Rolling Z-Score vs. Locked Formation σ:** `indicators.py` computes a rolling moving average and standard deviation (60-day window) during the trading phase. GGR (2006) computes the spread standard deviation (σ) *only once* over the 12-month In-Sample Formation period and rigidly *locks* this constant as the fixed threshold for the entire 6-month Out-of-Sample Trading period.

3. **Arbitrary Trade Triggers:** `strategies.py` sets arbitrary thresholds (`entry_z = 1.5`, `exit_z = 1.0`). GGR (2006) strictly defines entry at a 2σ **divergence** and requires the trade to be held until the spread exactly crosses **zero (perfect convergence)**.

4. **Isolated Pair Testing vs. Overlapping Portfolios:** `optimize.py` evaluates pairs in isolation using a 63-day roll. GGR (2006) constructs a unified Top 20 portfolio, rolls forward dynamically by exactly **1 month**, and concurrently manages **6 overlapping portfolios**, averaging their daily returns.

5. **Zero Frictions:** The current broker simulation applies 0% commission. A rigorous replication, especially for lower-liquidity markets like BIST, must incorporate explicit transaction costs (e.g., 10 bps per leg) to avoid overstating returns.

6. **Econometric Filters:** The current `pairs.py` utilizes Engle-Granger cointegration and Ornstein-Uhlenbeck half-life filters. While valid for modern applications, strict replication requires pairs to be sorted and selected purely by the SSD distance metric.

Part 2: The Corrected GGR (2006) Pipeline

Phase 1: Global Data Architecture & Exhaustive Screening

- **Data Ingestion:** Read the universe of tickers from `clean_turkish_tickers.txt`. Append the `.IS` suffix and download daily `Adjusted Close` and `Volume` from January 2010 to May 2026.
- **Zero-Volume Screen:** Construct a Boolean Volume Matrix. For any rolling 12-month Formation window, unconditionally reject any equity that records a daily volume of 0 or `NaN` on any trading day within that window.

Phase 2: Formation Period (12-Month In-Sample)

Operate on the dynamically filtered pool of highly liquid equities for the given 12-month window:

- **Normalization:** Divide the daily price series of each surviving stock by its initial price on Day 1 of the formation window: $P^*_{i,t} = P_{i,t}/P_{i,0}$.
- **SSD Computation:** Calculate the SSD between all mathematically possible combinations $N(N-1)/2$:

$$SSD_{A,B} = \sum_{t=1}^{252} \left(P^*_{A,t} - P^*_{B,t}\right)^2$$

- **Portfolio Construction:** Rank pairs by ascending SSD. Lock in the **Top 20** pairs.
- **Metric Locking:** Calculate the standard deviation (σ) of the normalized price spread ($P^*_{A,t} - P^*_{B,t}$) over these 12 months. Save this exact constant for the Trading phase.

Phase 3: Trading Engine (6-Month Out-of-Sample)

Step into the immediate 6-month Out-of-Sample window with the Top 20 pairs:

- **Execution Logic:** Do not recompute moving averages. Monitor the daily spread. If the spread expands beyond 2σ (using the locked formation σ), execute a dollar-neutral trade (Short \$1 of the overperforming leg, Long \$1 of the underperforming leg).
- **Exit Logic:**
  1. *Convergence:* Close the position immediately when the spread crosses 0.

2. *Time-Stop:* Force-close all open positions at the closing price on the final day of the 6-month window.

- **Frictions:** Deduct a realistic transaction cost (e.g., 10 basis points) per leg upon entry and exit.

Phase 4: Walk-Forward Overlapping Portfolios

- **Monthly Rolling Step:** Advance the 12-month Formation window and 6-month Trading window forward by exactly **1 month** iteratively through 2026.
- **Return Aggregation:** Because a new 6-month portfolio initiates monthly, manage up to 6 portfolios concurrently. The aggregate daily strategy return is the equal-weighted average of the daily returns generated by all active overlapping portfolios.

Phase 5: Capital Bases & Significance

Report daily percentage returns scaled by two distinct capital denominators:

- **Committed Capital:** Divide net profit by the total number of pair slots (20), accounting for the cash drag of unfunded pairs.
- **Fully Invested Capital:** Divide net profit only by the count of pairs actively holding an open trade.